Data Collection and Analysis Using SQL

Presented by: Hamidov Jeyhun

SQL-Based Analysis with Valentina Studio & PostgreSQL

Introduction

In this presentation, I will show how I collected and analyzed data using SQL.

You will learn about:

1. How I gathered the data
2. How I created data tables in SQL and imported the dataset
3. How I analyzed the data in SQL and drew meaningful conclusions

Without further delay, let’s get started!

Data Preparation

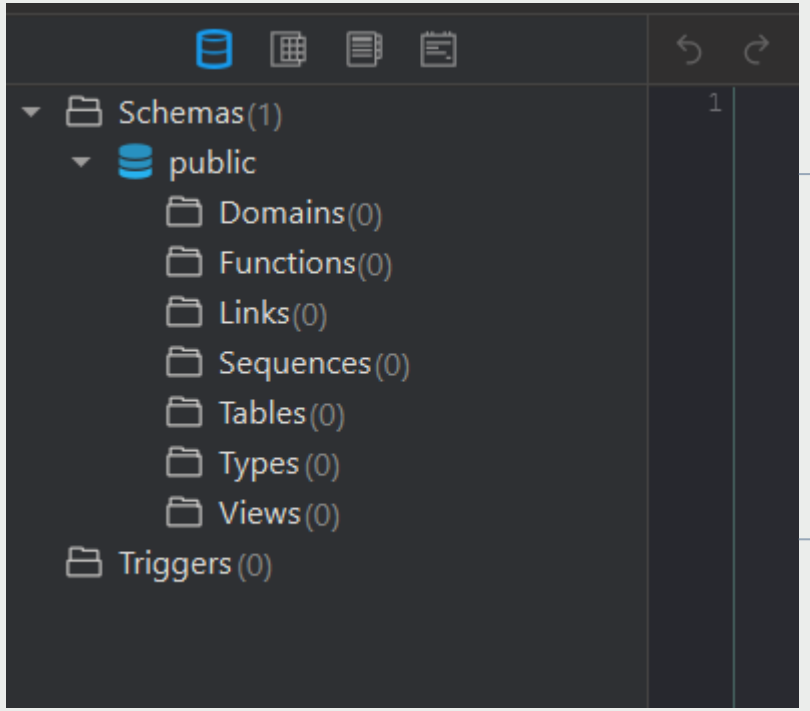
incident_systems	4/2/2026 9:42 PM	CSV File	6 KB
login_logs	4/2/2026 9:42 PM	CSV File	274 KB
network_events	4/2/2026 9:42 PM	CSV File	203 KB
organizations	4/2/2026 9:42 PM	CSV File	3 KB
security_incidents	4/2/2026 9:42 PM	CSV File	22 KB
systems	4/2/2026 9:42 PM	CSV File	15 KB
users	4/2/2026 9:42 PM	CSV File	31 KB

In this project, we already have a certain dataset.

However, since all of the data is in **CSV format**, it cannot be directly analyzed in SQL.

Therefore, we need to **import this data into the Valentina Studio program** to make it ready for analysis.

Creating Tables



As you can see here, there are no tables yet.

Therefore, we need to **write the SQL code for each table separately** and create them one by one.

After creating all the tables, we will be able to **perform our analyses on these tables**.

Let’s start to create all tables one by one

USERS TABLE

```sql
CREATE TABLE users(
    user_id VARCHAR(50) PRIMARY KEY,
    org_id VARCHAR(50),
    "role" VARCHAR(50)
)
```

The **Users** table contains information about individuals within organizations. It defines user roles and organizational affiliation, making it essential for analyzing user behavior, access control, and role-based security risks.

user_id → Unique identifier for each user

org_id → Identifier of the organization the user belongs to

role → Role of the user within the organization (e.g., Admin, Analyst, Employee)

LOGINS_LOG TABLE

```sql
CREATE TABLE logins_log(
    login_id VARCHAR(50) PRIMARY KEY,
    user_id VARCHAR(50),
    login_time date,
    ip_address VARCHAR(50),
    status VARCHAR(50) CHECK(status IN('Failed', 'Success'))
)
```

The **Logins_Log** table records user authentication activity. It is crucial for detecting suspicious behavior such as repeated failed logins, potential brute-force attacks, and unusual access patterns.

login_id → Unique identifier for each login attempt

user_id → Identifier of the user attempting to log in

login_time → Date of the login attempt

ip_address → IP address used during login

status → Outcome of the login attempt (Success or Failed)

NETWORKS_EVENT TABLE

```sql
CREATE TABLE networks_event(
    event_id VARCHAR(50) PRIMARY KEY,
    system_id VARCHAR(50),
    event_type VARCHAR(50),
    time_stamp DATE,
    severity VARCHAR(50)
)
```

The **Networks_Event** table stores system-level activity logs. It helps monitor system behavior and identify potential threats based on event type and severity.

event_id → Unique identifier for each system event

system_id → Identifier of the system where the event occurred

event_type → Type of event (e.g., Login, Network Scan, File Access)

time_stamp → Date when the event occurred

severity → Severity level of the event (e.g., Low, Medium, High, Critical)

ORGANIZATIONS TABLE

```sql
CREATE TABLE organizations(
    org_id VARCHAR(50) PRIMARY KEY,
    industry VARCHAR(50),
    country VARCHAR(50)
)
```

org_id → Unique identifier for each organization

industry → Industry sector of the organization (e.g., Finance, Tech, Healthcare)

country → Country where the organization operates

The **Organizations** table provides business context for the data. It enables analysis of how cybersecurity risks vary across industries and geographic locations.

SYSTEMS TABLE

```sql
CREATE TABLE systems(
    system_id VARCHAR(50) PRIMARY KEY,
    org_id VARCHAR(50),
    os_type VARCHAR(50),
    criticality VARCHAR(50)
)
```

The **Systems** table stores information about organizational infrastructure. It helps assess system-level risk and understand how different operating systems and critical systems are affected by security events.

system_id → Unique identifier for each system

org_id → Identifier of the organization that owns the system

os_type → Operating system used (e.g., Windows, Linux, Mac)

criticality → Importance level of the system (Low, Medium, High)

INCIDENT_SYSTEM TABLE

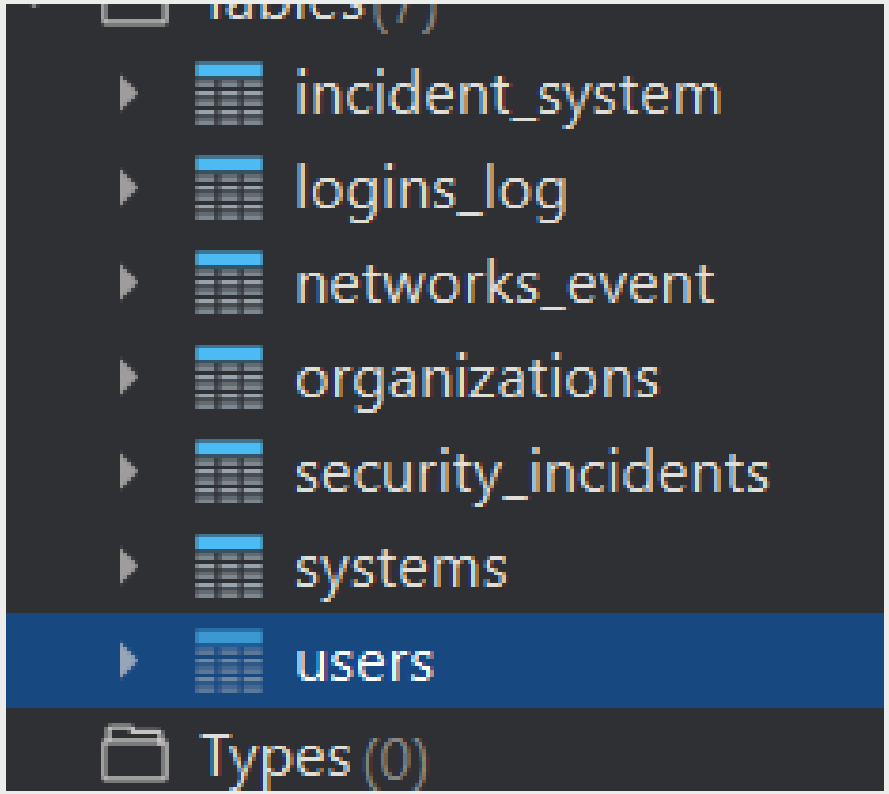
```sql
CREATE TABLE incident_system (
    incident_id VARCHAR(50) PRIMARY KEY,
    systems_id VARCHAR(50)
)
```

The **Incident_System** table links security incidents to specific systems. It plays a key role in identifying which systems are impacted by incidents and supports root cause analysis.

incident_id → Unique identifier for each incident

system_id → Identifier of the system associated with the incident

Database Setup Summary



In this part of the project, the dataset was collected and structured into a relational SQL database. Multiple tables were created with well-defined schemas, ensuring clear relationships between users, systems, organizations, events, and security incidents.

This structured design enables efficient querying, accurate analysis, and the ability to uncover meaningful cybersecurity insights in later stages of the project.

Dataset Overview

This dataset forms a robust cybersecurity monitoring framework that tracks user activity, system events, and security incidents across various organizations. By linking user login data, network telemetry, and incident records with organizational context, it provides a comprehensive view of the digital landscape. This integrated structure enables precise analysis of behavioral patterns and system vulnerabilities, transforming raw data into actionable insights for effective, scalable threat detection and strategic security decision-making.

Login Failure Analysis

```sql
SELECT
    status,
    count(*) AS total_attemps
FROM logins_log
GROUP BY status
```

	status	total_attempts
1	Failed	2960
2	Success	3040

The data reveals an unusually high login failure rate, accounting for nearly 50% of all attempts, which is significantly above normal system behavior. This pattern may indicate potential brute-force attacks, weak password practices among users, or insufficient authentication controls. Such a high proportion of failed logins increases the risk of unauthorized access attempts and suggests that the system may be exposed to security threats if not properly managed.

Top Failed Login Users Analysis

```sql
SELECT
    user_id,
    count( * ) AS failed attemps,
FROM logins_log
WHERE status = 'Failed'
GROUP BY user_id
ORDER BY failed_attemps DESC
LIMIT 10;
```

	user_id	failed_attempts
1	U0367	8
2	U1281	8
3	U0940	8
4	U1340	8
5	U0279	7
6	U1376	7
7	U0061	7
8	U1326	7
9	U1447	7
10	U0540	7

The failed login attempts are relatively evenly distributed across multiple users rather than being concentrated on a single account. This suggests that the issue is less likely to be a targeted brute-force attack on a specific user and more indicative of a broader pattern, such as multiple users struggling with authentication or a potential system-wide issue affecting login success rates. Additionally, the absence of extremely high failure counts for a single user reduces the likelihood of a focused attack but still indicates a systemic security or usability concern.

Failed Login by IP Analysis

```sql
SELECT
    ip_address,
    count( * ) AS failed attemps,
FROM logins_log
WHERE status = 'Failed'
GROUP BY ip_address
ORDER BY failed_attemps DESC
LIMIT 10;
```

	ip_address	failed_attempts
1	192.168.96.235	2
2	192.168.220.34	2
3	192.168.189.149	2
4	192.168.220.23	2
5	192.168.146.189	2
6	192.168.148.2	2
7	192.168.229.202	2
8	192.168.223.135	2
9	192.168.199.69	2
10	192.168.156.201	2

The failed login attempts are widely distributed across many different IP addresses, with no single IP showing a significantly higher number of failures. This strongly suggests that the system is not experiencing a traditional brute-force attack from a single source. Instead, the pattern may indicate either normal user behavior (such as occasional incorrect password entries) or a more distributed form of access attempts. Overall, the lack of concentration from specific IPs reduces the likelihood of targeted external attacks and points more toward a general usability or authentication issue.

Failed Login Attempts by User Role

```sql
SELECT
    u.role,
    count(l.login_id) AS failed_attemps
FROM logins_log l
JOIN users u ON l.user_id = u.user_id
WHERE l.status = 'Failed'
GROUP BY u.role
ORDER BY failed_attempts DESC
```

	role	failed_attempts
1	Admin	1008
2	Analyst	980
3	Employee	972

Failed login attempts are relatively evenly distributed across all user roles, with a slight concentration among Admin users. While the difference is not drastically large, the fact that Admin accounts have the highest number of failed attempts is a critical concern, as these accounts typically have elevated privileges and access to sensitive systems. This increases the potential risk, since even a small vulnerability in admin authentication could lead to severe security breaches. The overall distribution also reinforces the earlier finding that the issue is likely systemic rather than targeted, but the elevated exposure of Admin accounts makes them a key risk area.

Failed Login Attempts by Industry

```sql
SELECT
    o.industry,
    count(l.login_id) AS failed_attemps
FROM logins_log l
JOIN users u ON l.user_id = u.user_id
JOIN organizations o ON u.org_id = o.org_id
WHERE l.status = 'Failed'
GROUP BY o.industry
ORDER BY failed_attemps DESC
```

	industry	failed_attempts
1	Finance	846
2	Healthcare	811
3	Retail	743
4	Tech	560

The distribution of failed login attempts across industries reveals that the Finance and Healthcare sectors experience significantly higher authentication failures compared to others. This pattern is particularly important because these industries typically handle highly sensitive data, such as financial records and personal health information. The elevated number of failed attempts may indicate stricter security environments, higher user activity, or increased targeting by attackers. Combined with earlier findings, this suggests that while login failures are generally widespread, certain industries are inherently more exposed to risk due to the value of the data they manage.

Incidents by Operating System

```sql
SELECT
    s.os_type,
    count(si.incident_id) AS total_incidents
FROM security_incidents si
JOIN systems s ON si.org_id = s.org_id
GROUP BY s.os_type
ORDER BY total_incidents DESC
```

	os_type	total_incidents
1	Windows	905
2	Mac	854
3	Linux	746

The results indicate that Windows systems experience the highest number of security incidents; however, the difference between operating systems is not drastically large, suggesting that security risks are present across all platforms rather than being isolated to a single OS. The higher number for Windows may be influenced by its widespread usage in enterprise environments, making it a more frequent target for attacks. Overall, this pattern suggests that vulnerabilities are not OS-specific but rather related to broader security practices, system exposure, and usage patterns within organizations.

Top Industries by Security Incidents

```sql
SELECT
    o.org_id,
    o.industry,
    count(si.incident_id)AS total_incidents
FROM security_incidents AS si
JOIN organizations o ON si.org_id = o.org_id
GROUP BY o.org_id, o.industry
ORDER BY total_incidents DESC
LIMIT 10
```

	org_id	industry	total_incidents
1	ORG079	Finance	12
2	ORG082	Tech	11
3	ORG110	Retail	10
4	ORG078	Healthcare	9
5	ORG008	Finance	8
6	ORG107	Retail	8
7	ORG096	Retail	8
8	ORG053	Finance	8
9	ORG055	Tech	7
10	ORG058	Healthcare	7

The results indicate that security incidents are not concentrated within a single organization or industry but are instead distributed across multiple sectors, including Finance, Tech, and Retail. However, the presence of Finance organizations at the top highlights a potentially higher exposure to risk due to the sensitive nature of financial data. Additionally, the relatively close incident counts among top organizations suggest that cybersecurity threats are widespread and consistently affecting different types of businesses. This pattern reinforces the idea that security risks are systemic and not limited to specific companies or industries.

Yearly Login Trend Analysis

```sql
SELECT
    EXTRACT(YEAR FROM login_time) AS YEAR,
    status,
    count(*) AS total
FROM logins_log
GROUP BY YEAR, status
ORDER BY YEAR
```

	year	status	total
1	2023	Success	2225
2	2023	Failed	2132
3	2024	Failed	828
4	2024	Success	815

The data shows a significant decline in total login activity from 2023 to 2024, with both successful and failed attempts dropping sharply. However, despite the decrease in volume, the proportion of failed logins remains consistently high across both years, staying close to 50%. This suggests that the underlying issue affecting login success rates has not improved over time. The drop in total activity could be due to reduced system usage, fewer active users, or incomplete data for 2024, but the persistence of a high failure rate indicates a continuous systemic problem in authentication or user behavior.

Incident Type & Severity Analysis

```sql
SELECT
    incident_type,
    severity,
    count(*) AS total
FROM security_incidents
GROUP BY incident_type, severity
ORDER BY total DESC
```

	incident_type	severity	total
1	Data Breach	Low	53
2	Data Breach	Critical	49
3	Malware	High	49
4	Malware	Critical	48
5	Malware	Medium	45
6	Unauthorized Access	Critical	43
7	Malware	Low	43
8	Unauthorized Access	High	39
9	Unauthorized Access	Medium	34
10	Unauthorized Access	Low	33

The distribution of incidents across both type and severity reveals that threats are not only frequent but also occur at varying levels of impact. Notably, Data Breach incidents appearing simultaneously at low and critical levels may indicate inconsistent classification or varying scales of data exposure. Malware shows a balanced spread across all severity levels, suggesting it is a persistent and versatile threat affecting systems in different ways. Meanwhile, Unauthorized Access incidents being concentrated in higher severity levels highlights a significant security concern, as these incidents directly relate to breaches in access control. Overall, the data suggests that organizations are facing diverse and multi-layered cybersecurity threats, requiring flexible and well-prioritized security strategies.

Failed Logins vs Security Incidents

```sql
SELECT
    o.org_id,
    COUNT(DISTINCT l.login_id) AS failed_logins,
    COUNT(DISTINCT si.incident_id) AS incidents
FROM organizations o
LEFT JOIN users u ON o.org_id = u.org_id
LEFT JOIN logins_log l ON u.user_id = l.user_id AND l.status = 'Failed'
LEFT JOIN security_incidents si ON o.org_id = si.org_id
GROUP BY o.org_id;
```

The relationship between failed login attempts and security incidents appears to be inconsistent across organizations. While some organizations, such as ORG008 and ORG003, show a parallel increase in both failed logins and incidents—suggesting a possible link between authentication issues and security risks—others, like ORG002 and ORG006, demonstrate high login failures without a corresponding rise in incidents. This indicates that failed logins alone are not a definitive predictor of security incidents but may act as an early warning signal in certain contexts. The variation suggests that additional factors, such as security controls, system configurations, or incident detection capabilities, play a significant role in determining whether failed login activity escalates into actual security incidents.

	org_id	failed_logins	incidents
1	ORG001	17	7
2	ORG002	29	3
3	ORG003	27	7
4	ORG004	19	3
5	ORG005	19	1
6	ORG006	27	1
7	ORG007	16	3
8	ORG008	31	8
9	ORG009	23	4
10	ORG010	20	3

Final Conclusion

This project reveals that systemic authentication challenges, rather than targeted attacks, drive high login failure rates across all users and roles, with Admin accounts representing a critical security priority. Sector-specific analysis highlights that industries like Finance and Healthcare face heightened exposure, while incident distribution across various operating systems suggests that vulnerabilities are rooted in broader security practices rather than platform-specific flaws. Ultimately, while login failures may not always correlate directly with security breaches, they serve as essential early warning signals, necessitating a holistic cybersecurity strategy that integrates proactive user monitoring, robust system defenses, and rapid incident response to mitigate diverse threats like unauthorized access and data breaches.

This analysis helped us understand that cybersecurity risks are **systemic, multi-layered, and not limited to a single factor**. It showed the importance of combining user behavior analysis, system monitoring, and incident tracking to gain a complete picture of security risks and make better data-driven decisions.

Thank you for your attention!